

KESSLER

Low-Level Design (LLD)

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Traces to: KSL-HLD-001, KSL-SRS-001

1. Repository Layout

```
kessler/
├── documents/           SRS/HLD/LLD (src html + pdf), diagrams/, qa-report
├── backend/
│   ├── requirements.txt fastapi, uvicorn, pydantic-settings, sgp4, numpy, httpx, pytest, anthropo
│   ├── logs/           kessler.log (rotating, git-ignored)
│   └── app/
│       ├── main.py     app factory, middleware, handlers, startup pipeline
│       ├── core/       config.py · logging_config.py · exceptions.py
│       ├── models/     schemas.py (Pydantic v2 DTOs)
│       ├── logic/      cdm_parser.py · propagation.py · collision_probability.py
│       │               · screening.py · risk_engine.py                [pure, no I/O]
│       ├── service/    celestrak_client.py · spacetrack_client.py · cache_service.py
│       │               · conjunction_service.py · pipeline_service.py · insight_service.py
│       ├── api/        routes.py
│       └── data/       sample_gp.json · sample_cdms.json · cache.db (runtime)
├── backend/tests/      test_collision_probability.py · test_cdm_parser.py
│                       · test_screening.py · test_risk_engine.py · test_api.py
└── ui/                Vite + React + TS (src/api, src/components, src/pages, src/styles)
```

2. Core (app/core)

2.1 config.py

```
class Settings(BaseSettings):
    # env prefix KESSLER_, .env supported
    app_name: str = "KESSLER"
    demo_mode: bool | None = None # None → auto: demo iff no spacetrack creds
    spacetrack_user: str | None = None
    spacetrack_pass: str | None = None
    anthropic_api_key: str | None = None
    data_dir: Path = <backend>/app/data
    log_dir: Path = <backend>/logs
    gp_ttl_seconds: int = 3600 # CON-1: never refetch GP inside 1 h
    cdm_ttl_seconds: int = 1800
    screening_max_window_hours: int = 72
    default_hbr_m: float = 20.0
    settings = Settings() # imported everywhere; test-overridable
```

2.2 exceptions.py

Class	code	HTTP	Raised by
KesslerError(base)	KESSLER_ERROR	500	—
ConfigError	CONFIG_ERROR	500	core/service init
DataSourceError	DATA_SOURCE_ERROR	502	celestrak/spacetrack clients
RateLimitError	RATE_LIMITED	503	spacetrack throttle guard
CdmParseError	CDM_PARSE_ERROR	422	logic/cdm_parser
PropagationError	PROPAGATION_ERROR	500	logic/propagation, screening
NotFoundError	NOT_FOUND	404	services/API

Global FastAPI handlers convert `KesslerError` → envelope `{"error": {"code", "message", "detail", "request_id"}}`; a catch-all handler maps any other exception to 500 `INTERNAL_ERROR` after logging with stack trace (NFR-1).

2.3 logging_config.py

```
FORMAT = "%(asctime)s | %(levelname)-7s | %(name)s | rid=%(request_id)s | %(message)s"
setup_logging(): console handler (INFO) + RotatingFileHandler(logs/kessler.log, 5MB, 3 backups)
RequestIdFilter injects request_id (contextvar set by middleware; "-" outside requests)
```

3. Logic Layer (app/logic) — algorithms

3.1 cdm_parser.py

```
@dataclass CdmObject: designator, name, object_type, r_rtn: vec3|None,
                        v_rtn: vec3|None, cov_rtn: 3x3|None      # metres, m/s, m²
@dataclass Cdm: cdm_id, created, tca: datetime, miss_distance_m, relative_speed_ms,
                sat1: CdmObject, sat2: CdmObject, pc_reported: float|None,
                emergency_reportable: bool, source: str, raw: dict

parse_cdm_kvn(text: str) -> Cdm          # CCSDS 508.0-B-1 KVN: header → relative
                                         # metadata → OBJECT1/OBJECT2 blocks (state+cov)
parse_cdm_public_row(row: dict) -> Cdm    # Space-Track cdm_public JSON row
                                         # (CDM_ID, CREATED, TCA, MIN_RNG, PC,
                                         # SAT_1_ID, SAT_1_NAME, SAT1_OBJECT_TYPE, ...)
```

Validation: TCA parseable and TZ-aware, both designators present, numeric fields finite, covariance symmetric-PSD when supplied (symmetrised; non-PSD → flagged, dropped to max-Pc path). All violations accumulated then raised in one `CdmParseError(detail=[ ... ]) (FR-CDM-3)`.

3.2 propagation.py

```
load_satrec(omm: dict) -> Satrec          # sgp4.exporter-compatible OMM load
state_teme(sat, t: datetime) -> (r_km, v_kms) # raises PropagationError on sgp4 code≠0
sample_range(satA, satB, t0, t1, step_s) -> ndarray # vectorised |rA-rB| over window
refine_tca(satA, satB, t_guess, half_window_s=90) -> (tca, miss_m, rel_speed)
# golden-section minimisation of range(t), tol 0.25 s → then quadratic fit for sub-s TCA
```

3.3 collision_probability.py — the mathematical core (FR-PC)

```
encounter_plane_projection(r_rel, v_rel, cov1, cov2) -> (mu_2d, C_2d)
# 1. combined covariance C = cov1 + cov2 (independent errors)
# 2. basis: ez = v_rel/|v_rel|; ex = r_rel - (r_rel·ez)ez, normalised; ey = ez × ex
# 3. project miss vector and C into (ex, ey) plane → mu (2,), C_2d (2×2)

pc_foster(mu, C, hbr_m, n_r=60, n_theta=90) -> float
# numerical polar integration of bivariate Gaussian over disc radius=hbr
#  $P_c = \iint N(x; \mu, C) dA$ , midpoint rule in (r,  $\theta$ ); relative err < 1e-3 for n defaults

pc_chan(mu, C, hbr_m, terms=20) -> float
# Chan (2008) equivalent-area series: transform to isotropic space,
#  $u = hbr^2/(ox \cdot oy \dots)$ ,  $v = miss^2 \dots$ ;  $P_c = e^{\{-v/2\}} \sum (v/2)^m/m! \cdot [1 - e^{\{-u/2\}} \sum \dots]$ 

pc_max(miss_m, hbr_m) -> float
# covariance-free conservative bound (Alfano style):
# maximise isotropic- $\sigma$   $P_c$  over  $\sigma \rightarrow \sigma^* = miss/\sqrt{2} \rightarrow P_{c\_max} = (hbr^2/miss^2) \cdot e^{\{-1\}}$ 
# (clipped to 1; documented as triage-grade upper bound)

compute_pc(cdm, hbr_m) -> PcResult(pc, method, cross_check, divergence_flag, pc_type)
# dispatch: covariance → foster + chan cross-check ( $|\Delta|/\max > 5\% \Rightarrow \text{flag}$ )
# no covariance → pc_max, pc_type="max"
```

3.4 screening.py

```
screen(asset_omm, catalog: list[omm], window_h, threshold_km=10) -> list[CloseApproach]
Step 1 altitude sieve: skip object if [perigee, apogee] bands are > threshold apart
Step 2 coarse scan: shared time grid (60 s), vectorised ranges; local minima < 5×threshold
Step 3 refine each candidate with refine_tca(); keep miss ≤ threshold
Step 4 annotate RTN components of miss vector, rel speed, pc_max
```

3.5 risk_engine.py

```
classify(pc, pc_type, miss_m, cov_available) -> RiskClass # FR-RSK-1 thresholds
urgency(risk, tca, created) -> float 0..100
# base by class (CRITICAL 70) + up to 20 for time-to-TCA < 24 h + up to 10 for stale data
recommend(risk, tca, cov_available) -> str # deterministic action text
```

4. Data Design — SQLite DDL

```
CREATE TABLE IF NOT EXISTS kv_cache (  
  key TEXT PRIMARY KEY, payload TEXT NOT NULL,          -- JSON  
  fetched_at TEXT NOT NULL, ttl_seconds INTEGER NOT NULL);  
CREATE TABLE IF NOT EXISTS events (  
  event_id TEXT PRIMARY KEY, tca TEXT NOT NULL, risk TEXT NOT NULL,  
  urgency REAL NOT NULL, document TEXT NOT NULL,        -- enriched event JSON  
  updated_at TEXT NOT NULL);  
CREATE INDEX IF NOT EXISTS ix_events_urgency ON events(urgency DESC);  
CREATE TABLE IF NOT EXISTS pipeline_runs (  
  id INTEGER PRIMARY KEY AUTOINCREMENT, agent TEXT NOT NULL,  
  started_at TEXT NOT NULL, finished_at TEXT, status TEXT NOT NULL,  
  items INTEGER DEFAULT 0, error TEXT);
```

5. API Contract (app/api/routes.py)

Method & Path	Query/body	200 response (summary)
GET /api/v1/health	—	{status, version, data_mode, uptime_s}
GET /api/v1/conjunctions	limit≤100, min_risk	{data_mode, generated_at, events: [EventSummary]} sorted by urgency
GET /api/v1/conjunctions/{event_id}	—	EventDetail: cdm, pc{value,method,cross_check,divergence,type}, risk, urgency, action, geometry{r_rtn, miss_rtn}
GET /api/v1/conjunctions/{event_id}/insight	—	{briefing, source:"ai" "template"}
GET /api/v1/satellites/{norad}/track	minutes≤360, step_s≥10	{norad, name, points:[{t, lat, lon, alt_km, r_eci_km}]}
POST /api/v1/screening/run	{norad_id, window_hours≤72, threshold_km}	{asset, window, candidates:[CloseApproach]}
GET /api/v1/pipeline/status	—	{agents: [{id,name,status,last_run,duration_ms,items,error}]}
GET /api/v1/catalog	q, limit	[[norad, name, object_type, apogee_km, perigee_km, inclination_deg]]

422 validation errors are re-shaped into the standard envelope by an override of FastAPI's `RequestValidationError` handler; every response carries `X-Request-Id` .

6. Service Layer Detail

6.1 spacetrack_client.py — rate-limit governor

```
class SpaceTrackClient:
    MIN_INTERVAL_S = 3.0          # ≥ 3 s between calls ⇒ < 20/min (CON-1 margin)
    HOURLY_BUDGET = 250           # < 300/hr with margin
    async login()                 # POST /ajaxauth/login, session cookie
    async fetch_cdm_public(limit=200) -> list[dict]
        # /basicspacedata/query/class/cdm_public/orderby/TCA asc/limit/{n}/format/json
    async fetch_gp(norad_ids)      # only if cache-miss AND ttl expired (guarded by cache_service)
    # every call: acquire throttle lock → check budget → httpx call (timeout 30 s,
    # 2 retries exp backoff) → on 401 re-login once → errors → DataSourceError/RateLimitError
```

6.2 conjunction_service.py — pipeline orchestration (sequence)

```
refresh_pipeline():                # startup + scheduled
    1 pipeline.start("catalog_sync") → gp = cache.get_or_fetch("gp:active", loader)
    2 pipeline.start("cdm_ingest")   → raw = cache.get_or_fetch("cdm:latest", loader|demo)
    3 pipeline.start("parse")        → cdms = [parse_cdm_public_row(r)] ; quarantine failures
    4 pipeline.start("pc_engine")    → for each: compute_pc(cdm, settings.default_hbr_m)
    5 pipeline.start("risk")         → classify + urgency + recommend (+ insight lazily)
    6 pipeline.start("publish")      → upsert events table; snapshot generated_at
    each stage: try/except → pipeline.finish(status, items, error); stage failure uses
    last-known-good data and marks agent "degraded" – pipeline never aborts the app.
```

6.3 insight_service.py

If `anthropic_api_key` set: Claude Messages API call (short, temperature 0.2, max_tokens 400) with event JSON context → analyst briefing; 10 s timeout; on any exception log WARN and return template briefing (`source: "template"`). Never raises to caller (FR-RSK-4).

7. UI Design (ui/)

Module	Detail
src/api/client.ts	fetch wrapper: JSON, error-envelope unwrap → typed <code>ApiError</code> ; abortable; single retry on 5xx.
src/types.ts	TS mirrors of Pydantic DTOs (EventSummary, EventDetail, PipelineAgent, TrackPoint...).
components/Globe3D.tsx	R3F canvas: Earth sphere (custom gradient material + atmosphere rim), star-free deep background, orbit polylines (satellite.js propagation client-side, 1 orbit period, 256 segments), TCA marker + miss-vector line for selected event; OrbitControls (damped).
components/EventList.tsx	Urgency-sorted table: risk chip, Pc (scientific), miss km, rel speed km/s, TCA countdown; row select → focus.
components/EventDetail.tsx	Panel: objects, Pc block (method, cross-check, divergence flag, pc_type), RTN miss components bar, action, AI/template briefing.
pages/PipelinePage.tsx	Embeds <code>public/diagrams/pipeline-flow.html</code> (iframe) and polls <code>/pipeline/status</code> every 5 s, posting status into the iframe (postMessage) to light up nodes.
styles/	Design tokens: bg #05070d, panel rgba(13,18,32,.85), accent cyan #59d8ff, alert amber #ffb454, critical #ff5d5d, mono numerals (font-variant-numeric: tabular-nums). No decorative patterns (FR-UI-4).

8. Test Plan

Suite	Cases
test_collision_probability	isotropic closed-form vs Foster ($<1e-3$ rel); Foster vs Chan agreement on reference geometries; max-Pc bound $\geq$ both methods for matched miss/HBR; degenerate covariance → max-Pc fallback; Pc→0 for huge miss; Pc→~1 for tiny miss with tight covariance.
test_cdm_parser	KVN fixture round-trip; cdm_public row fixture; missing TCA/designator → CdmParseError listing both; non-PSD covariance flagged.
test_screening	synthetic co-planar pair with engineered close pass detected within 1 s of truth; altitude sieve excludes disjoint orbit; empty-catalogue no-op.
test_risk_engine	threshold boundaries (1e-4, 1e-5, 1e-7); miss<1 km escalation without covariance; urgency monotonic in Pc and time-to-TCA.
test_api	TestClient: /health; /conjunctions contract + sort; unknown event → 404 envelope; screening validation (window > 72 → 422 envelope); pipeline status shape.

